=== src/m3/answers_1.html === <section class="cover"> Real Analysis II · Semester 5 · Module 3 <h1>Answer key: The Riemann Integral</h1> Formal solutions to all 40 questions of the Module 3 model paper (Bartle & Sherbert §7.1–§7.3).   Before you read an answer Each answer starts with Know first: topic codes from File 1 (Prepare). If one feels new, read that topic first, try writing the skeleton yourself, and then read the solution. Grey "Marks" notes show a typical split (a study guide, not an official scheme): statement · setup · key step · conclusion. </section>
<section class="page"> Section A · Short answers<small>3 marks each. Ceiling 48.</small>
1PYQ MTS-26 Q4also MAT-22 Q3, MAT-21 Q5, MAT-22 MCQ 16 Definition of a partition and its norm; norms on [0,4] and [2,9]. Know firstT01M02 Definition · Partition and Norm (Bartle §7.1, eq. (1))If I := [a,b] is a closed bounded interval in \mathbb{R}, a partition of I is a finite, ordered set P := (x_0, x_1, \dots, x_{n-1}, x_n) of points in I such that a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b. The norm (or mesh) of P is the length of the largest subinterval I_i = [x_{i-1},x_i]:
\lVert P\rVert := \max\{x_1 - x_0,\, x_2 - x_1,\, \dots,\, x_n - x_{n-1}\}.
 SkeletonList \Delta x_i = x_i - x_{i-1} for each partition; take \lVert P\rVert = \max_i \Delta x_i. On [0,4]: <ul> P_1 = (0,1,2,4): subinterval lengths are 1-0=1, 2-1=1, 4-2=2, so \lVert P_1\rVert = \max\{1,1,2\} = 2. P_2 = (0,2,3,4): subinterval lengths are 2, 1, 1, so \lVert P_2\rVert = \max\{2,1,1\} = 2. P_3 = (0,1,1.5,2,3.4,4): subinterval lengths are 1, 0.5, 0.5, 1.4, 0.6, so \lVert P_3\rVert = \max\{1,0.5,0.5,1.4,0.6\} = 1.4. On [2,9] (MAT-22 MCQ 16): \lVert Q_1\rVert = \max\{2,2,2,1\} = 2; \lVert Q_2\rVert = \max\{4,3\} = 4; \lVert Q_3\rVert = \max\{3,2,2\} = 3. Thus Q_2 = (2,6,9) has the largest norm (MCQ option B). \lVert P_1\rVert = 2,\ \lVert P_2\rVert = 2,\ \lVert P_3\rVert = 1.4; largest on [2,9] is Q_2 with \lVert Q_2\rVert = 4. Marks: definitions 1 · norms on [0,4] 1 · largest norm on [2,9] 1.
2PYQ MTS-22 Q4 Tagged partition and Riemann sum; f(x) = x^2 on [0,4] with P = (0,1,2,4). Know firstT02P02M03 Definition · Tagged Partition and Riemann Sum (Bartle §7.1, eq. (2))If a point t_i \in I_i = [x_{i-1},x_i] is selected from each subinterval of a partition P for i=1,\dots,n, the points t_i are called tags and \dot P := \{([x_{i-1},x_i],t_i)\}_{i=1}^n is a tagged partition of [a,b]. For f : [a,b] \to \mathbb{R}, the Riemann sum of f corresponding to \dot P is
S(f;\dot P) := \sum_{i=1}^n f(t_i)(x_i - x_{i-1}).
 SkeletonWrite the subintervals [0,1], [1,2], [2,4] with widths \Delta x_1 = 1, \Delta x_2 = 1, \Delta x_3 = 2, evaluate f(t_i) = t_i^2, and sum f(t_1)\Delta x_1 + f(t_2)\Delta x_2 + f(t_3)\Delta x_3. Left-endpoint tags t_1 = 0, t_2 = 1, t_3 = 2:
S(f;\dot P_{\text{left}}) = 0^2(1 - 0) + 1^2(2 - 1) + 2^2(4 - 2) = 0 + 1 + 8 = 9.
 Right-endpoint tags t_1 = 1, t_2 = 2, t_3 = 4:

S(f;\dot P_{\text{right}}) = 1^2(1 - 0) + 2^2(2 - 1) + 4^2(4 - 2) = 1 + 4 + 32 = 37.
 S(f;\dot P_{\text{left}}) = 9 and S(f;\dot P_{\text{right}}) = 37. Marks: definition 1 · left-endpoint sum 1 · right-endpoint sum 1. </section> <section class="page"> 3PYQ MAT-21 MCQ 9 · adapted Definition of f \in \mathcal{R}[a,b]; why \int_a^b f is not an ordinary function limit. Know firstT03M02M03 Definition 7.1.1 · Riemann Integrable (Bartle p. 201)A function f : [a,b] \to \mathbb{R} is said to be Riemann integrable on [a,b] if there exists a number L \in \mathbb{R} such that for every \varepsilon > 0 there exists \delta_\varepsilon > 0 such that if \dot P is any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon, then \lvert S(f;\dot P) - L\rvert < \varepsilon.
The set of all Riemann integrable functions on [a,b] is denoted by \mathcal{R}[a,b], and we write L = \int_a^b f or \int_a^b f(x)\,dx. Why this is not an ordinary function limit as \lVert P\rVert \to 0 (Remark 7.1.1): For a given positive number h = \lVert P\rVert, there are infinitely many partitions P with norm h and infinitely many choices of tags t_i \in I_i for each such P. Because many different Riemann sums S(f;\dot P) correspond to the same number \lVert P\rVert, S(f;\dot P) is not a single-valued function of \lVert P\rVert. (In MAT-21 MCQ 9, the function satisfying Definition 7.1.1 is called Riemann integrable, option A.) Marks: Definition 7.1.1 word for word 1.5 · notation \mathcal{R}[a,b] and \int_a^b f 0.5 · non-function limit explanation 1.
4 Uniqueness of the Riemann integral (Theorem 7.1.2). Know firstT04P05M08 SkeletonAssume L' and L'' both satisfy Definition 7.1.1. Given \varepsilon > 0, take \delta'_{\varepsilon/2} and \delta''_{\varepsilon/2}, put \delta_\varepsilon = \min\{\delta'_{\varepsilon/2},\delta''_{\varepsilon/2}\}, and pick one \dot P with \lVert P\rVert < \delta_\varepsilon to get \lvert L' - L''\rvert < \varepsilon. Proof of Theorem 7.1.2 (Bartle p. 201) Suppose L' and L'' both satisfy Definition 7.1.1 for f \in \mathcal{R}[a,b], and let \varepsilon > 0 be arbitrary.We will show \lvert L' - L''\rvert < \varepsilon for every \varepsilon > 0 (P05). By Definition 7.1.1 with tolerance \varepsilon/2 > 0, there exists \delta'_{\varepsilon/2} > 0 such that \lVert P_1\rVert < \delta'_{\varepsilon/2} \Rightarrow \lvert S(f;\dot P_1) - L'\rvert < \varepsilon/2, and there exists \delta''_{\varepsilon/2} > 0 such that \lVert P_2\rVert < \delta''_{\varepsilon/2} \Rightarrow \lvert S(f;\dot P_2) - L''\rvert < \varepsilon/2.Definition 7.1.1 applied to L' and L''. Put \delta_\varepsilon := \min\{\delta'_{\varepsilon/2}, \delta''_{\varepsilon/2}\} > 0, and choose a tagged partition \dot P of [a,b] with \lVert P\rVert < \delta_\varepsilon. Then both inequalities of Step 2 hold for \dot P, so by the Triangle Inequality, \lvert L' - L''\rvert \le \lvert L' - S(f;\dot P)\rvert + \lvert S(f;\dot P) - L''\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
Since \varepsilon > 0 is arbitrary, L' = L''. ▌Common \delta_\varepsilon argument (M08), Triangle Inequality, and arbitrary \varepsilon rule (P05).   Marks: setup with \varepsilon/2 1 · \delta_\varepsilon = \min\{\delta'_{\varepsilon/2},\delta''_{\varepsilon/2}\} and triangle inequality 1.5 · conclusion 0.5.

5PYQ MTS-25 Q4 Constant function f(x) = k is in \mathcal{R}[a,b] with \int_a^b k = k(b-a). Know firstT05T03M01 SkeletonCompute S(f;\dot P) = k(b-a) for every tagged partition \dot P; given \varepsilon > 0, take \delta_\varepsilon = 1. Proof (Bartle Example 7.1.4(a), p. 202) Let f(x) := k for all x \in [a,b]. If \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n is any tagged partition of [a,b], then
S(f;\dot P) = \sum_{i=1}^n f(t_i)(x_i - x_{i-1}) = \sum_{i=1}^n k(x_i - x_{i-1}) = k\sum_{i=1}^n (x_i - x_{i-1}) = k(x_n - x_0) = k(b - a).
Factor out k (P02) and telescope \sum_{i=1}^n (x_i - x_{i-1}) = b - a (M01). Hence for any \varepsilon > 0, we can choose \delta_\varepsilon := 1 > 0 so that if \dot P has \lVert P\rVert < \delta_\varepsilon, then \lvert S(f;\dot P) - k(b-a)\rvert = \lvert k(b-a) - k(b-a)\rvert = 0 < \varepsilon.
Since \varepsilon > 0 is arbitrary, f \in \mathcal{R}[a,b] and \int_a^b k = k(b-a). ∎Definition 7.1.1 satisfied with L = k(b-a) and \delta_\varepsilon = 1.   \displaystyle\int_a^b k = k(b - a). Marks: telescoping Riemann sum S(f;\dot P) = k(b-a) 1.5 · explicit \delta_\varepsilon = 1 and \lvert S - k(b-a)\rvert = 0 < \varepsilon 1 · conclusion 0.5. </section>
<section class="page"> 6PYQ MAT-22 MCQ 18 · adaptedalso MAT-21 MCQ 13, MTS-22 MCQ 7 · adapted Changing g \in \mathcal{R}[a,b] at finitely many points (Theorem 7.1.3); \int_0^2 h; finite vs countable exception sets. Know firstT06M03M04 SkeletonFor one exceptional point c, bound \lvert S(f;\dot P) - S(g;\dot P)\rvert \le 2(\lvert f(c)\rvert + \lvert g(c)\rvert)\lVert P\rVert < \varepsilon/2 when \lVert P\rVert < \delta_1, and combine with \lvert S(g;\dot P) - L\rvert < \varepsilon/2. Proof of Theorem 7.1.3 (Bartle pp. 201–202) It suffices to prove the case of one exceptional point c \in [a,b] (finitely many points follow by induction). Let L := \int_a^b g and suppose f(x) = g(x) for all x \ne c.Induction reduces m exceptional points to 1 point. For any tagged partition \dot P, the sums S(f;\dot P) and S(g;\dot P) agree except on at most two subintervals (when c is a partition point tag of two adjacent intervals). Thus \lvert S(f;\dot P) - S(g;\dot P)\rvert \le 2\big(\lvert f(c)\rvert + \lvert g(c)\rvert\big)\lVert P\rVert.
At most 2 terms differ (M03), each of width \le \lVert P\rVert (M04). Given \varepsilon > 0, choose \delta_1 > 0 with \delta_1 < \varepsilon/\big(4(\lvert f(c)\rvert + \lvert g(c)\rvert + 1)\big) and choose \delta_2 > 0 so that \lVert P\rVert < \delta_2 \Rightarrow \lvert S(g;\dot P) - L\rvert < \varepsilon/2. Let \delta := \min\{\delta_1,\delta_2\} > 0. For \lVert P\rVert < \delta,
\lvert S(f;\dot P) - L\rvert \le \lvert S(f;\dot P) - S(g;\dot P)\rvert + \lvert S(g;\dot P) - L\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,
so f \in \mathcal{R}[a,b] and \int_a^b f = L = \int_a^b g. ∎Triangle Inequality and Definition 7.1.1.
Applications & MCQ resolutions: <ul> For h(x) = 2 on [0,1), h(1) = 3, h(x) = 1 on (1,2]: h differs from the step function g = 2\varphi_{[0,1]} + 1\varphi_{(1,2]} only at c = 1, so

\int_0^2 h = \int_0^2 g = 2(1) + 1(1) = 3. If f(x) = 0 except at finitely many points c_1,\dots,c_n (MAT-22 MCQ 18, MAT-21 MCQ 13): comparing with g \equiv 0 gives both f \in \mathcal{R}[a,b] (option A) and \int_a^b f = 0 (option B). (Note: in MAT-22 MCQ 18 both A and B are true; in MAT-21 MCQ 13 option B is the unique correct option.) Ambiguous past question note (MTS-22 MCQ 7): When f(x) = g(x) off the countably infinite set \{1/n : n \in \mathbb{N}\}, knowing f \in \mathcal{R}[0,1] does not guarantee g \in \mathcal{R}[0,1] in general (e.g. f = 0 and g(1/n) = n is unbounded, so g \notin \mathcal{R}[0,1]); if both f, g \in \mathcal{R}[0,1] then \int_0^1 f = \int_0^1 g since f - g = 0 outside a null set. The intended exam key for MTS-22 MCQ 7 treats the unquantified statement (C) as the false option, or assumes boundedness. \displaystyle\int_0^2 h = 3; if f=0 off a finite set, f\in\mathcal{R}[a,b] and \int_a^b f = 0. Marks: proof of 7.1.3 1.5 · \int_0^2 h = 3 0.5 · finite vs countable discussion 1.

7PYQ MAT-22 MCQ 15 · adaptedalso MAT-21 MCQ 11, 17, MTS-22 MCQ 8, 9 · adapted Boundedness Theorem 7.1.6; H(1/k) = k not integrable; converses. Know firstT08P05 Theorem 7.1.6 · Boundedness Theorem (Bartle p. 205)If f \in \mathcal{R}[a,b], then f is bounded on [a,b]. SkeletonSuppose f \in \mathcal{R}[a,b] is unbounded. Fix \delta > 0 for \varepsilon = 1 so \lvert S(f;\dot P)\rvert < \lvert L\rvert + 1. Fix a partition Q with \lVert Q\rVert < \delta; f is unbounded on some I_k; pick t_k \in I_k huge to contradict \lvert S(f;\dot Q)\rvert < \lvert L\rvert + 1. Proof of Theorem 7.1.6 (Bartle pp. 205–206) Suppose f \in \mathcal{R}[a,b] with integral L, and assume for contradiction that f is unbounded on [a,b]. Taking \varepsilon = 1 in Definition 7.1.1 gives \delta > 0 such that \lVert P\rVert < \delta \Rightarrow \lvert S(f;\dot P) - L\rvert < 1, whence \lvert S(f;\dot P)\rvert < \lvert L\rvert + 1.Definition 7.1.1 with \varepsilon = 1 and Triangle Inequality. Let Q = \{[x_{i-1},x_i]\}_{i=1}^n be a partition of [a,b] with \lVert Q\rVert < \delta. Since f is unbounded on [a,b], it must be unbounded on at least one subinterval [x_{k-1},x_k].If f were bounded on each of the n pieces by M_i, it would be bounded on [a,b] by \max_i M_i. Tag Q by setting t_i := x_i for i \ne k, and choose t_k \in [x_{k-1},x_k] so that
\lvert f(t_k)(x_k - x_{k-1})\rvert > \lvert L\rvert + 1 + \left\lvert\sum_{i\ne k} f(t_i)(x_i - x_{i-1})\right\rvert.
By the Reverse Triangle Inequality, \lvert S(f;\dot Q)\rvert \ge \lvert f(t_k)(x_k - x_{k-1})\rvert - \big\lvert\sum_{i\ne k} f(t_i)(x_i - x_{i-1})\big\rvert > \lvert L\rvert + 1, contradicting Step 1. Thus f is bounded. ∎Reverse Triangle Inequality (P05).   Applications & Converses: <ul> For H(1/k) = k (k \in \mathbb{N}), H(x) = 0 elsewhere on [0,1]: since H(1/k) = k \to \infty, H is unbounded on [0,1], so by Theorem 7.1.6, H \notin \mathcal{R}[0,1] (MTS-22 MCQ 9 option A; MAT-21 MCQ 17 option B; MTS-22 MCQ 8 option D). f \in \mathcal{R}[a,b] does not imply f is continuous or monotone (Thomae's function C8 is integrable but neither continuous nor monotone; MAT-22 MCQ 15 and MAT-21 MCQ 11 option A). Not every bounded function is integrable: Dirichlet's function C7 is bounded (0 \le f \le 1) but f \notin \mathcal{R}[0,1]. Marks: statement & proof of 7.1.6 2 · H \notin \mathcal{R}[0,1] and converses 1. </section>

<section class="page"> 8PYQ MAT-21 MCQ 10 · adapted Elementary step function \varphi_J (Lemma 7.2.4) and \int_0^4 g. Know firstT13T06M10 SkeletonFor J = [c,d] and \lVert P\rVert < \delta, trap S(\varphi_{[c,d]};\dot P) between (d-c)-2\delta and (d-c)+2\delta and pick \delta_\varepsilon = \varepsilon/4; extend to (c,d), [c,d), (c,d] via Theorem 7.1.3. Proof of Lemma 7.2.4 (Bartle p. 210) First let J = [c,d] \subseteq [a,b] (c < d). For any tagged partition \dot P with \lVert P\rVert < \delta, the union U_1 of subintervals in \dot P with tag t_i \in [c,d] satisfies [c+\delta, d-\delta] \subseteq U_1 \subseteq [c-\delta, d+\delta] (M10), so (d - c) - 2\delta \le S(\varphi_{[c,d]};\dot P) \le (d - c) + 2\delta.Tag-inclusion bounds (M10 / Exercise 7.1.13). Given \varepsilon > 0, take \delta_\varepsilon := \varepsilon/4 > 0; then \lVert P\rVert < \delta_\varepsilon \Rightarrow \lvert S(\varphi_{[c,d]};\dot P) - (d - c)\rvert \le 2(\varepsilon/4) = \varepsilon/2 < \varepsilon. Thus \varphi_{[c,d]} \in \mathcal{R}[a,b] and \int_a^b \varphi_{[c,d]} = d - c.Definition 7.1.1 with \delta_\varepsilon = \varepsilon/4. The other three intervals (c,d), [c,d), (c,d] have indicator functions differing from \varphi_{[c,d]} at at most the two endpoints c, d. By Theorem 7.1.3, each \varphi_J \in \mathcal{R}[a,b] with \int_a^b \varphi_J = d - c. ∎Theorem 7.1.3 (T06).   Evaluation of \int_0^4 g (MAT-21 MCQ 10): Since g(x) = 3 on [0,2] and g(x) = 2 on (2,4], we write g = 3\varphi_{[0,2]} + 2\varphi_{(2,4]}. By Lemma 7.2.4 and Theorem 7.1.5, \int_0^4 g = 3(2 - 0) + 2(4 - 2) = 6 + 4 = 10 (MCQ option C). \displaystyle\int_a^b \varphi_J = d - c and \displaystyle\int_0^4 g = 10. Marks: proof for [c,d] 1.5 · extension by 7.1.3 0.5 · \int_0^4 g = 10 1.
9 Oriented integrals (Definition 7.2.12, Theorem 7.2.13) and check for f(x) = x, \alpha = 3, \beta = 1, \gamma = 0. Know firstT19T17 Definition 7.2.12 & Theorem 7.2.13 (Bartle p. 214)If f \in \mathcal{R}[a,b] and \alpha, \beta \in [a,b] with \alpha < \beta, we define
$$\int_\beta^\alpha f := -\int_\alpha^\beta f \qquad \text{and} \qquad \int_\alpha^\alpha f := 0.$$
Theorem 7.2.13: For any numbers \alpha, \beta, \gamma \in [a,b], \displaystyle\int_\alpha^\beta f = \int_\alpha^\gamma f + \int_\gamma^\beta f. Verification for f(x) = x with \alpha = 3, \beta = 1, \gamma = 0: Using \int_u^v x\,dx = \tfrac12(v^2 - u^2) for u < v: <ul> \displaystyle\int_3^1 x\,dx = -\int_1^3 x\,dx = -\tfrac12(3^2 - 1^2) = -4. \displaystyle\int_3^0 x\,dx + \int_0^1 x\,dx = -\int_0^3 x\,dx + \int_0^1 x\,dx = -\tfrac12(9 - 0) + \tfrac12(1 - 0) = -\frac92 + \frac12 = -4. Both sides equal -4. Marks: Definition 7.2.12 1 · Theorem 7.2.13 statement 0.5 · verification 1.5.
10 Why FTC 1 fails for H(x) = 2\sqrt x on [0,b] and K(x) = x^2\cos(1/x^2) on [0,1]. Know firstT21T20T08 Theorem 7.3.1 (FTC 1) requires three hypotheses: (a) F continuous on [a,b]; (b) F'(x) = f(x) on [a,b]\setminus E for a finite set E; and (c) f \in \mathcal{R}[a,b]. In both examples, hypothesis (c) fails because the derivative is unbounded (Theorem 7.1.6): <ul> For H(x) = 2\sqrt x on [0,b] (Example

7.3.2(d)): H is continuous on [0,b] and H'(x) = \frac{1}{\sqrt x} on (0,b) (E = \{0\}). However, h(x) = \frac{1}{\sqrt x} is unbounded on (0,b) as x \to 0^+, so by Theorem 7.1.6, h \notin \mathcal{R}[0,b] regardless of how h(0) is defined. For K(x) = x^2\cos(1/x^2) (x \in (0,1]), K(0) = 0 on [0,1] (Example 7.3.2(e), C19): For x > 0, the Product and Chain Rules give K'(x) = 2x\cos(1/x^2) + \frac{2}{x}\sin(1/x^2), and at x = 0, K'(0) = \lim_{x\to 0}\frac{x^2\cos(1/x^2)}{x} = 0. Thus K is continuous and differentiable at every point of [0,1] (E = \varnothing). Yet along x_m = (\frac{\pi}{2} + 2m\pi)^{-1/2} \to 0^+, \sin(1/x_m^2) = 1 and K'(x_m) = \frac{2}{x_m} \to \infty, so K' is unbounded on [0,1] and K' \notin \mathcal{R}[0,1]. Marks: identify hypothesis (c) f\in\mathcal{R} via 7.1.6 1 · H'(x)=1/\sqrt x unbounded 1 · K'(x) formula, K'(0)=0, and unboundedness 1. </section>
<section class="page"> 11PYQ MAT-21 Q6 Definition of the indefinite integral F(z) = \int_a^z f; proof that F is Lipschitz and continuous (Theorem 7.3.4). Know firstT22T17T07 Definition 7.3.3 · Indefinite Integral (Bartle p. 218)If f \in \mathcal{R}[a,b], the function defined by
F(z) := \int_a^z f \quad \text{for } z \in [a,b]
is called the indefinite integral of f with basepoint a. Proof of Theorem 7.3.4 (Bartle p. 218) Let w, z \in [a,b] with w \le z. By the Additivity Theorem 7.2.9, F(z) = \int_a^z f = \int_a^w f + \int_w^z f = F(w) + \int_w^z f, so F(z) - F(w) = \int_w^z f.Additivity Theorem 7.2.9 (T17). Since \lvert f(x)\rvert \le M on [a,b], we have -M \le f(x) \le M on [w,z]. By Theorem 7.1.5(c) and Example 7.1.4(a),
-M(z - w) \le \int_w^z f \le M(z - w),
whence \lvert F(z) - F(w)\rvert = \left\lvert\int_w^z f\right\rvert \le M\lvert z - w\rvert for all z, w \in [a,b].Monotonicity of the integral (7.1.5c) and constant integral (7.1.4a). Thus F is Lipschitz on [a,b] with constant M, and hence uniformly continuous and continuous on [a,b] (given \varepsilon > 0, take \delta := \varepsilon/(M+1)). ∎Lipschitz \Rightarrow continuous (P09).   Marks: Definition 7.3.3 1 · Additivity step 1 · Lipschitz bound and continuity 1.
12 Indefinite integrals of \operatorname{sgn} x on [-1,1] and Thomae's h on [0,1] (Examples 7.3.7(a),(b)). Know firstT24T23T09 1. For f(x) = \operatorname{sgn} x on [-1,1] with basepoint -1 (Example 7.3.7(a), C20): <ul> If -1 \le x < 0, then f(t) = -1 on [-1,x), so F(x) = \int_{-1}^x (-1)\,dt = -(x + 1) = \lvert x\rvert - 1. If 0 \le x \le 1, Additivity gives F(x) = \int_{-1}^0 (-1)\,dt + \int_0^x 1\,dt = -1 + x = \lvert x\rvert - 1. Thus F(x) = \lvert x\rvert - 1 for all x \in [-1,1]. Since F'_-(0) = -1 \ne +1 = F'_+(0), F'(0) does not exist, so F is not an antiderivative of f on [-1,1]. 2. For Thomae's function h on [0,1] (Example 7.3.7(b), C8): Since \int_0^1 h = 0 and h(x) \ge 0, we have 0 \le H(x) = \int_0^x h \le \int_0^1 h = 0, so H(x) = 0 for all x \in [0,1]. Although H'(x) = 0 exists everywhere on [0,1], at every rational x = m/n \in \mathbb{Q}\cap[0,1] we have H'(x) = 0 \ne 1/n = h(x), so H is not an antiderivative of h on [0,1]. F(x) =

\lvert x\rvert - 1 (F'(0) DNE); H(x) = 0 (H'(x) \ne h(x) on \mathbb{Q}\cap[0,1]). Neither is an antiderivative. Marks: F(x) = \lvert x\rvert - 1 on [-1,1] 1 · F'(0) DNE 1 · Thomae H \equiv 0 and H' \ne h on rationals 1.
13PYQ MTS-25 Q7also MTS-26 Q7, MTS-22 MCQ 10 Definition of a null set; proof that \mathbb{Q}\cap[0,1] is null; properties of null sets. Know firstT26P03P13M07 Definition 7.3.10(a) · Null Set (Bartle p. 220)A set Z \subseteq \mathbb{R} is said to be a null set if for every \varepsilon > 0 there exists a countable collection \{(a_k,b_k)\}_{k=1}^\infty of open intervals such that
Z \subseteq \bigcup_{k=1}^\infty (a_k,b_k) \qquad \text{and} \qquad \sum_{k=1}^\infty (b_k - a_k) \le \varepsilon.
 Proof that \mathbb{Q}_1 = \mathbb{Q}\cap[0,1] is a null set (Example 7.3.11, p. 221) Since \mathbb{Q}_1 = \mathbb{Q}\cap[0,1] is countable, enumerate its elements as \mathbb{Q}_1 = \{r_1, r_2, \dots, r_k, \dots\}.Countability of \mathbb{Q}\cap[0,1] (P13). Given \varepsilon > 0, for each k \in \mathbb{N} define the open interval J_k := \left(r_k - \dfrac{\varepsilon}{2^{k+1}},\, r_k + \dfrac{\varepsilon}{2^{k+1}}\right), which contains r_k and has length \dfrac{\varepsilon}{2^k}.Geometric halving trick (M07). Then \mathbb{Q}_1 \subseteq \bigcup_{k=1}^\infty J_k and the total length is \sum_{k=1}^\infty \dfrac{\varepsilon}{2^k} = \varepsilon\sum_{k=1}^\infty \dfrac{1}{2^k} = \varepsilon. Since \varepsilon > 0 is arbitrary, \mathbb{Q}\cap[0,1] is a null set. ∎Geometric series \sum_{k=1}^\infty 1/2^k = 1 (P03).   Properties (MTS-22 MCQ 10): Any subset of a null set is null (A true); the union of two null sets is null (B true); every countable set is null by the same proof (C true). However, not every null set is countable (D is the false statement): the Cantor set is an uncountable null set. Marks: Definition 7.3.10(a) 1 · proof for \mathbb{Q}\cap[0,1] 1.5 · properties & Cantor set remark 0.5.
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<section class="page"> 14PYQ MAT-22 MCQ 27, 28, 29 · adaptedalso MAT-21 MCQ 19, 20 · adapted FTC 1 evaluations: \int_a^b f = F(b)-F(a), \int_{-2}^3 g = 5/2, and \int_a^b \lvert x\rvert\,dx = B(b)-B(a). Know firstT20T21 (a) MAT-22 MCQ 27 / MAT-21 MCQ 19: If f is continuous on [a,b] (hence f \in \mathcal{R}[a,b] by 7.2.7) and F is any antiderivative of f on [a,b] (F is continuous by 6.1.2 and F' = f with E = \varnothing), then by Theorem 7.3.1, \int_a^b f(x)\,dx = F(b) - F(a) (MCQ option B).
(b) Bartle Ex 7.3.3 (MAT-22 MCQ 28 / MAT-21 MCQ 20): On [-2,3], G(x) = \tfrac12\lvert x^2 - 1\rvert is continuous. For \lvert x\rvert > 1, G(x) = \tfrac12(x^2-1) has G'(x) = x = g(x); for \lvert x\rvert < 1, G(x) = \tfrac12(1-x^2) has G'(x) = -x = g(x). Thus G'(x) = g(x) off the finite set E = \{-1, 1\}, and g \in \mathcal{R}[-2,3] (piecewise continuous). By FTC 1, \int_{-2}^3 g(x)\,dx = G(3) - G(-2) = \frac12\lvert 9 - 1\rvert - \frac12\lvert 4 - 1\rvert = 4 - \frac32 = \frac52 \quad \text{(MCQ option B)}.
 (c) Bartle Ex 7.3.4 (MAT-22 MCQ 29): For B(x) = -\tfrac12 x^2 (x < 0), \tfrac12 x^2 (x \ge 0), we have B'(x) = -x = \lvert x\rvert for x < 0, B'(x) = x = \lvert x\rvert for x > 0, and

B'_-(0) = 0 = B'_+(0) = \lvert 0\rvert. Thus B is continuous and B'(x) = \lvert x\rvert on all of [a,b] with E = \varnothing, and \lvert x\rvert \in \mathcal{R}[a,b] (continuous). By FTC 1, \int_a^b \lvert x\rvert\,dx = B(b) - B(a) (MCQ option B). (a) F(b)-F(a); (b) \displaystyle\int_{-2}^3 g = \frac52 (E=\{-1,1\}); (c) \displaystyle\int_a^b \lvert x\rvert\,dx = B(b)-B(a) (E=\varnothing). Marks: part (a) 0.5 · part (b) with E=\{-1,1\} and value 5/2 1.5 · part (c) with E=\varnothing 1.

15PYQ MAT-22 Q35 Evaluate \int_0^1 x^2(1 - \sqrt x)\,dx using the Fundamental Theorem of Calculus. Know firstT20T21T15 SkeletonWrite f(x) = x^2 - x^{5/2} and F(x) = \tfrac13 x^3 - \tfrac27 x^{7/2} on [0,1]. Check hypotheses (a), (b), (c) of Theorem 7.3.1 with E = \varnothing, then compute F(1) - F(0). Solution On [0,1], expand the integrand as f(x) := x^2(1 - \sqrt x) = x^2 - x^{5/2}. Since f is continuous on the closed bounded interval [0,1], Theorem 7.2.7 implies that f \in \mathcal{R}[0,1] (hypothesis (c) of Theorem 7.3.1).Continuous \Rightarrow integrable (Theorem 7.2.7, T15). Define F : [0,1] \to \mathbb{R} by
F(x) := \frac{x^3}{3} - \frac{2}{7}x^{7/2}.
Then F is continuous on [0,1] (hypothesis (a)), and F'(x) = x^2 - x^{5/2} = f(x) for all x \in [0,1] (hypothesis (b) with E = \varnothing).Power rule for derivatives on [0,1]. By the First Form of the Fundamental Theorem of Calculus (Theorem 7.3.1),
\int_0^1 x^2(1 - \sqrt x)\,dx = F(1) - F(0) = \left(\frac{1}{3} - \frac{2}{7}\right) - 0 = \frac{7 - 6}{21} = \frac{1}{21}. \quad \blacksquare
Theorem 7.3.1 (T20).   \displaystyle\int_0^1 x^2(1 - \sqrt x)\,dx = \frac{1}{21}. Marks: f \in \mathcal{R}[0,1] by continuity 1 · antiderivative F(x) = \tfrac13 x^3 - \tfrac27 x^{7/2} 1 · F(1)-F(0) = 1/21 1.

16PYQ MAT-22 MCQ 19, 26 · adapted Trigonometric symmetry \int_0^{\pi/2} f(\cos x)\,dx and translation \int_{a+c}^{b+c} f(y-c)\,dy = \int_a^b f. Know firstT25T19 (a) MAT-22 MCQ 26: Let \varphi(x) := \tfrac{\pi}{2} - x on [0,\pi/2]. Since \varphi'(x) = -1 is continuous and \cos x = \sin(\tfrac{\pi}{2} - x), Theorem 7.3.8 and Definition 7.2.12 give \int_0^{\pi/2} f(\cos x)\,dx = -\int_0^{\pi/2} f(\sin(\varphi(x)))\,\varphi'(x)\,dx = -\int_{\pi/2}^0 f(\sin u)\,du = \int_0^{\pi/2} f(\sin u)\,du.
Next, on [\pi/2,\pi], substituting u = \psi(x) := \pi - x (\psi' = -1, \sin x = \sin(\pi - x)) gives \int_{\pi/2}^\pi f(\sin x)\,dx = \int_0^{\pi/2} f(\sin u)\,du. By Additivity 7.2.9, \int_0^\pi f(\sin x)\,dx = 2\int_0^{\pi/2} f(\sin x)\,dx, so \int_0^{\pi/2} f(\cos x)\,dx = \tfrac12\int_0^\pi f(\sin x)\,dx. Thus options (A) and (B) are equal to \int_0^{\pi/2} f(\cos x)\,dx, while (C) \int_0^\pi f(\sin x)\,dx is not equal to it (twice as large whenever the integral is non-zero). (b) Bartle Ex 7.1.15 / MAT-22 MCQ 19: For g(y) = f(y - c) on [a+c,b+c], any tagged partition \dot P_c = \{([y_{i-1},y_i],s_i)\} of [a+c,b+c] shifts by -c to a tagged partition \dot P = \{([y_{i-1}-c,y_i-c],s_i-c)\} of [a,b] with the same norm and S(g;\dot P_c) = S(f;\dot P). Thus g \in \mathcal{R}[a+c,b+c] and \int_{a+c}^{b+c} g = \int_a^b f (MCQ option B; when f is continuous this also follows from Theorem 7.3.8 with \varphi(y) = y - c, \varphi'(y) = 1). <div

class="final">(a) Not equal to (C) \int_0^\pi f(\sin x)\,dx; (b) \displaystyle\int_{a+c}^{b+c} g = \int_a^b f (option B). Marks: \int_0^{\pi/2} f(\cos x)\,dx = \int_0^{\pi/2} f(\sin x)\,dx 1 · \tfrac12\int_0^\pi f(\sin x)\,dx 1 · translate formula 1. </section> <section class="page"> 17PYQ MAT-22 MCQ 30 · adapted If f : [0,1] \to \mathbb{R} is continuous and \int_0^x f = \int_x^1 f for all x \in [0,1], then f \equiv 0. Know firstT23T22T17T15 SkeletonSet F(x) = \int_0^x f; use Additivity to write \int_x^1 f = F(1) - F(x), deduce F(x) = \tfrac12 F(1) is constant, and differentiate via Theorem 7.3.6. Proof (Bartle Exercise 7.3.16) Since f is continuous on [0,1], Theorem 7.2.7 implies that f \in \mathcal{R}[0,1]. Define the indefinite integral F(x) := \int_0^x f for x \in [0,1].Theorem 7.2.7 (T15) proves (B) f \in \mathcal{R}[0,1]. By the Additivity Theorem 7.2.9, F(1) = \int_0^1 f = \int_0^x f + \int_x^1 f. Since \int_0^x f = \int_x^1 f by hypothesis, we have 2F(x) = F(1), so F(x) = \tfrac12 F(1) is constant on [0,1] (in fact, setting x = 0 gives F(1) = 2F(0) = 0, so F(x) = 0 for all x \in [0,1]).Additivity Theorem 7.2.9 (T17). Since f is continuous on [0,1], Theorem 7.3.6 (FTC 2) gives f(x) = F'(x) = 0 for all x \in [0,1]. ∎Theorem 7.3.6 (T23). (In MAT-22 MCQ 30, both (A) f(x)=0\ \forall x and (B) f\in\mathcal{R}[0,1] hold, so the answer is (C) "All the above".)   Marks: f \in \mathcal{R}[0,1] 0.5 · 2F(x) = F(1) \Rightarrow F \equiv 0 via Additivity 1.5 · f = F' = 0 via FTC 2 1. 18PYQ MAT-22 MCQ 23 · adapted If f is continuous on [a,b], f \ge 0, and \int_a^b f = 0, then f(x) = 0 for all x \in [a,b]. Know firstT15T13T07 SkeletonSuppose f(c) > 0 for some c \in [a,b]. Use continuity at c with \varepsilon = \tfrac12 f(c) to find a closed subinterval J \subseteq [a,b] of length d-c > 0 where f \ge \tfrac12 f(c), getting \int_a^b f \ge \tfrac12 f(c)(d-c) > 0. Proof (Bartle Exercises 7.2.8–7.2.9) Suppose for contradiction that there is some c \in [a,b] with f(c) > 0. Since f is continuous at c, taking \varepsilon := \tfrac12 f(c) > 0 gives \delta > 0 such that if x \in [a,b] and \lvert x - c\rvert < \delta, then f(x) > f(c) - \tfrac12 f(c) = \tfrac12 f(c).Definition of continuity at c with \varepsilon = \tfrac12 f(c). Choose a non-degenerate closed subinterval J = [c_1, d_1] \subseteq [a,b] \cap (c-\delta, c+\delta) with c_1 < d_1. Since f(x) \ge 0 on [a,b] and f(x) > \tfrac12 f(c) on J, we have \tfrac12 f(c)\varphi_J(x) \le f(x) for all x \in [a,b].Lower bound by an elementary step function (M06). By Lemma 7.2.4 and Theorem 7.1.5(c), \int_a^b f \ge \int_a^b \tfrac12 f(c)\varphi_J = \tfrac12 f(c)(d_1 - c_1) > 0, which contradicts \int_a^b f = 0. Therefore f(x) = 0 for all x \in [a,b] (MAT-22 MCQ 23 option A). ∎Monotonicity 7.1.5(c) and Lemma 7.2.4 (T13).   Why continuity cannot be dropped (Ex 7.2.9): Define f(a) := 1 and f(x) := 0 for x \in (a,b]. Then f(x) \ge 0 on [a,b] and \int_a^b f = 0 by Theorem 7.1.3, yet f(a) = 1 \ne 0. Marks: continuity neighborhood with f > \tfrac12 f(c) 1 · step-function lower bound \int_a^b f > 0 1 · counterexample without continuity 1.

19PYQ MTS-22 Q14 Leibniz rule for variable-limit integrals; derivatives of \int_0^{x^2}(1+t^3)^{-1}\,dt and \int_{x^2}^x \sqrt{1+t^2}\,dt. Know firstT23T19P11 Leibniz rule for variable limits (Bartle Ex 7.3.10; PYQ MTS-22 Q14): Let f : [a,b] \to \mathbb{R} be continuous and let u, v : [c,d] \to [a,b] be differentiable. If I(z) := \int_a^z f(t)\,dt, then I'(z) = f(z) by Theorem 7.3.6 (FTC 2). Since G(x) = \int_{u(x)}^{v(x)} f(t)\,dt = I(v(x)) - I(u(x)) by Theorem 7.2.13, the Chain Rule 6.1.6 gives
G'(x) = I'(v(x))\,v'(x) - I'(u(x))\,u'(x) = f(v(x))\,v'(x) - f(u(x))\,u'(x).
 (a) F(x) = \int_0^{x^2} (1 + t^3)^{-1}\,dt (Bartle Ex 7.3.11(a)): Here f(t) = (1 + t^3)^{-1} is continuous on [0,1], u(x) = 0, and v(x) = x^2 (v'(x) = 2x), so
F'(x) = \frac{1}{1 + (x^2)^3}\cdot 2x = \frac{2x}{1 + x^6}.
 (b) F(x) = \int_{x^2}^x \sqrt{1 + t^2}\,dt (Bartle Ex 7.3.11(b)): Here f(t) = \sqrt{1 + t^2} is continuous on [0,1], v(x) = x (v'=1), and u(x) = x^2 (u'=2x), so
F'(x) = \sqrt{1 + x^2}\cdot 1 - \sqrt{1 + (x^2)^2}\cdot 2x = \sqrt{1 + x^2} - 2x\sqrt{1 + x^4}.
 (a) F'(x) = \dfrac{2x}{1 + x^6}; (b) F'(x) = \sqrt{1 + x^2} - 2x\sqrt{1 + x^4}. Marks: general rule via FTC 2 + Chain Rule 1 · part (a) 1 · part (b) 1.
20 Mean Value Theorem for Integrals (Bartle Exercise 7.2.16). Know firstT15T07P07 Proof (Bartle Exercise 7.2.16) Since f is continuous on the closed bounded interval [a,b], f \in \mathcal{R}[a,b] (Theorem 7.2.7) and by the Maximum–Minimum Theorem 5.3.4 (P07) f attains its minimum m := \min_{[a,b]} f and maximum M := \max_{[a,b]} f on [a,b].Theorem 7.2.7 and Max–Min Theorem 5.3.4 (P07). Since m \le f(x) \le M for all x \in [a,b], Theorem 7.1.5(c) and Example 7.1.4(a) give
m(b - a) \le \int_a^b f \le M(b - a), \quad \text{so} \quad m \le \frac{1}{b - a}\int_a^b f \le M.
Monotonicity of the integral (7.1.5c) and division by b - a > 0. By Bolzano's Intermediate Value Theorem 5.3.7 (P07), f attains every value between m and M, so there exists c \in [a,b] such that f(c) = \frac{1}{b-a}\int_a^b f, i.e. \int_a^b f = f(c)(b - a). ∎Bolzano's IVT 5.3.7. (Alternative: apply the MVT 6.2.4 to F(x) = \int_a^x f, which gives c \in (a,b).)   Fails for step functions: On [0,2], g(x) = 1 on [0,1) and g(x) = 2 on [1,2] has \int_0^2 g = 3, but g(c)(2 - 0) = 3 \iff g(c) = 3/2, which g never attains. Marks: m(b-a) \le \int_a^b f \le M(b-a) 1 · IVT conclusion 1 · step-function counterexample 1.
</section>
=== src/m3/answers_2.html === <section class="page"> Section B · Paragraph / problem answers (Part 1: Q21–Q28)<small>6 marks each. Ceiling 72.</small>
21PYQ MAT-21 Q29also MTS-22 Q5, MTS-25 Q5, MAT-22 MCQ 14 Linearity \int_a^b (f+g) = \int_a^b f + \int_a^b g and monotonicity f \le g \Rightarrow \int_a^b f \le \int_a^b g (Theorem 7.1.5(b),(c)). Know firstT07T03P02P05M08 Theorem 7.1.5(b), (c) (Bartle

p. 204)Suppose f and g are in \mathcal{R}[a,b]. Then: (b) The function f + g is in \mathcal{R}[a,b] and \int_a^b (f+g) = \int_a^b f + \int_a^b g. (c) If f(x) \le g(x) for all x \in [a,b], then \int_a^b f \le \int_a^b g. SkeletonCheck S(f+g;\dot P) = S(f;\dot P) + S(g;\dot P) and f \le g \Rightarrow S(f;\dot P) \le S(g;\dot P). Given \varepsilon > 0, choose \delta_\varepsilon = \min\{\delta_1,\delta_2\} so both sums are within \varepsilon/2 of their integrals. Use Triangle Inequality for (b) and sandwiching for (c). Proof of Theorem 7.1.5(b), (c) (Bartle pp. 204–205) For any tagged partition \dot P = \{([x_{i-1},x_i],t_i)\}_{i=1}^n of [a,b], properties of finite sums (P02) yield
$$S(f+g;\dot P) = \sum_{i=1}^n \big(f(t_i) + g(t_i)\big)(x_i - x_{i-1}) = S(f;\dot P) + S(g;\dot P),$$
and if f(x) \le g(x) on [a,b], then since x_i - x_{i-1} > 0 we have f(t_i)(x_i - x_{i-1}) \le g(t_i)(x_i - x_{i-1}), so S(f;\dot P) \le S(g;\dot P).Additivity and order preservation of finite sums (P02). Let \varepsilon > 0 be given. Since f, g \in \mathcal{R}[a,b], Definition 7.1.1 provides \delta_1 > 0 and \delta_2 > 0 such that
$$\lVert P\rVert < \delta_1 \implies \left\lvert S(f;\dot P) - \int_a^b f\right\rvert < \frac{\varepsilon}{2}, \qquad \lVert P\rVert < \delta_2 \implies \left\lvert S(g;\dot P) - \int_a^b g\right\rvert < \frac{\varepsilon}{2}.$$
Define \delta_\varepsilon := \min\{\delta_1, \delta_2\} > 0, and let \dot P be any tagged partition of [a,b] with \lVert P\rVert < \delta_\varepsilon.Definition 7.1.1 with tolerance \varepsilon/2 and common \delta_\varepsilon (M08). Proof of (b): By Step 1, Step 2, and the Triangle Inequality,
$$\left\lvert S(f+g;\dot P) - \left(\int_a^b f + \int_a^b g\right)\right\rvert = \left\lvert\left(S(f;\dot P) - \int_a^b f\right) + \left(S(g;\dot P) - \int_a^b g\right)\right\rvert \le \left\lvert S(f;\dot P) - \int_a^b f\right\rvert + \left\lvert S(g;\dot P) - \int_a^b g\right\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$
Since \varepsilon > 0 is arbitrary, f + g \in \mathcal{R}[a,b] and \int_a^b (f+g) = \int_a^b f + \int_a^b g.Triangle Inequality (P05) and Definition 7.1.1. Proof of (c): For the same \dot P with \lVert P\rVert < \delta_\varepsilon, unpacking the two absolute value inequalities of Step 2 gives
$$\int_a^b f - \frac{\varepsilon}{2} < S(f;\dot P) \quad \text{and} \quad S(g;\dot P) < \int_a^b g + \frac{\varepsilon}{2}.$$
Combining these with S(f;\dot P) \le S(g;\dot P) from Step 1 gives \int_a^b f - \frac{\varepsilon}{2} < \int_a^b g + \frac{\varepsilon}{2}, or \int_a^b f < \int_a^b g + \varepsilon. Since \varepsilon > 0 is arbitrary, \int_a^b f \le \int_a^b g. ∎Arbitrary \varepsilon rule (P05).   Why strict inequality < need not hold (MAT-22 MCQ 14): Option (B) \int_a^b f \le \int_a^b g is the true statement; strict < fails whenever f = g, or when f \equiv 0 and g(a) = 1, g(x) = 0 on (a,b] (where \int_a^b f = 0 = \int_a^b g). Marks: Riemann sum relations 1 · common \delta_\varepsilon setup 1 · proof of (b) 2 · proof of (c) and MCQ remark 2. </section> <section class="page"> 22 Thomae's function h on [0,1]: definition, proof that h \in \mathcal{R}[0,1] with \int_0^1 h = 0, and the set E for h(x) \ge 1/5. Know firstT09T03P14M03M10 Definition of Thomae's Function (Bartle Example 7.1.7, p. 206)Define h : [0,1] \to \mathbb{R} by h(x) := 0 if x \in [0,1] is irrational, h(0) := 1, and h(x) := 1/n if x \in [0,1] is the rational number x = m/n where m, n

\in \mathbb{N} have no common integer factor except 1. SkeletonGiven \varepsilon > 0, let E = \{x \in [0,1] : h(x) \ge \varepsilon/2\}, which has finitely many points n. Choose \delta = \varepsilon/(4n). Split \dot P into \dot P_1 (tags in E, at most 2n subintervals) and \dot P_2 (tags outside E, where h < \varepsilon/2). Proof that h \in \mathcal{R}[0,1] and \int_0^1 h = 0 (Bartle p. 206) Let \varepsilon > 0 be given. Since h(m/n) = 1/n \ge \varepsilon/2 \iff n \le 2/\varepsilon, there are only finitely many denominators n \in \mathbb{N} and finitely many numerators m \in \{1,\dots,n\} satisfying 1/n \ge \varepsilon/2. Thus the set E := \{x \in [0,1] : h(x) \ge \varepsilon/2\} is a finite set.Archimedean property (P13). Let n be the number of points in E, and define \delta := \dfrac{\varepsilon}{4n} > 0. Let \dot P be any tagged partition of [0,1] with \lVert P\rVert < \delta. Separate \dot P into \dot P_1 (the subset of subintervals with tags in E) and \dot P_2 (the subset with tags elsewhere in [0,1]).Tag-splitting tool (M10). Allowing for a point of E serving as an endpoint tag of two adjacent subintervals, \dot P_1 has at most 2n subintervals, of total length at most 2n\delta = 2n\cdot\frac{\varepsilon}{4n} = \frac{\varepsilon}{2}. Since 0 < h(t_i) \le 1 on \dot P_1, we have
S(h;\dot P_1) \le 1 \cdot 2n\delta = \frac{\varepsilon}{2}.
Endpoint tag count (M03) and h(x) \le 1. For tags t_i in \dot P_2, we have t_i \notin E, so 0 \le h(t_i) < \varepsilon/2. Since the total length of the subintervals in \dot P_2 is at most 1 (the length of [0,1]), we have
S(h;\dot P_2) < \frac{\varepsilon}{2}\cdot 1 = \frac{\varepsilon}{2}.
Height < \varepsilon/2 times total width \le 1 (M04). Adding Steps 3 and 4 gives
0 \le S(h;\dot P) = S(h;\dot P_1) + S(h;\dot P_2) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
Since \varepsilon > 0 is arbitrary, h \in \mathcal{R}[0,1] and \int_0^1 h = 0. ∎Definition 7.1.1 with L = 0.   Explicit list of E = \{x \in [0,1] : h(x) \ge 1/5\}: We collect 0 (where h(0)=1) and all fractions m/n \in (0,1] in lowest terms with denominator n \in \{1,2,3,4,5\}: E = \left\{0,\, 1,\, \frac12,\, \frac13,\, \frac23,\, \frac14,\, \frac34,\, \frac15,\, \frac25,\, \frac35,\, \frac45\right\} (11 points); \displaystyle\int_0^1 h = 0. Marks: definition of h 1 · finiteness of E and \delta = \varepsilon/(4n) 1.5 · estimates on \dot P_1 and \dot P_2 2.5 · list of 11 points in E 1.
23PYQ MAT-21 Q8 State and prove the Cauchy Criterion for Riemann integrability (Theorem 7.2.1). Know firstT10T03P06 Theorem 7.2.1 · Cauchy Criterion (Bartle p. 208)A function f : [a,b] \to \mathbb{R} belongs to \mathcal{R}[a,b] if and only if for every \varepsilon > 0 there exists \eta_\varepsilon > 0 such that if \dot P and \dot Q are any tagged partitions of [a,b] with \lVert P\rVert < \eta_\varepsilon and \lVert Q\rVert < \eta_\varepsilon, then
\lvert S(f;\dot P) - S(f;\dot Q)\rvert < \varepsilon.
 Skeleton(\Rightarrow) Put \eta_\varepsilon = \delta_{\varepsilon/2} and use the Triangle Inequality. (\Leftarrow) Take decreasing \delta_n > 0 for \varepsilon = 1/n, choose \dot P_n with \lVert P_n\rVert < \delta_n, show (S(f;\dot P_m)) is Cauchy in \mathbb{R} with limit A, let m \to \infty to get \lvert S(f;\dot

P_n) - A\rvert \le 1/n, and pick K > 2/\varepsilon. Proof of Theorem 7.2.1 (Bartle p. 208) (\Rightarrow) Suppose f \in \mathcal{R}[a,b] with integral L. Given \varepsilon > 0, let \eta_\varepsilon := \delta_{\varepsilon/2} > 0 from Definition 7.1.1. If \dot P, \dot Q have \lVert P\rVert < \eta_\varepsilon and \lVert Q\rVert < \eta_\varepsilon, then \lvert S(f;\dot P) - L\rvert < \varepsilon/2 and \lvert S(f;\dot Q) - L\rvert < \varepsilon/2, so \lvert S(f;\dot P) - S(f;\dot Q)\rvert \le \lvert S(f;\dot P) - L\rvert + \lvert L - S(f;\dot Q)\rvert < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
Definition 7.1.1 and Triangle Inequality (P05). (\Leftarrow) For each n \in \mathbb{N}, let \delta_n > 0 be the number \eta_{1/n} such that \lVert P\rVert, \lVert Q\rVert < \delta_n \Rightarrow \lvert S(f;\dot P) - S(f;\dot Q)\rvert < 1/n. By replacing \delta_n with \min\{\delta_1,\dots,\delta_n\} if necessary, we may assume \delta_n \ge \delta_{n+1} for all n \in \mathbb{N}.Hypothesis with \varepsilon = 1/n; forcing \delta_n decreasing ensures \lVert P_m\rVert < \delta_m \le \delta_n for m > n. For each n \in \mathbb{N}, choose a tagged partition \dot P_n with \lVert P_n\rVert < \delta_n. If m > n, then both \dot P_m and \dot P_n have norm <\delta_n, so \lvert S(f;\dot P_m) - S(f;\dot P_n)\rvert < \frac{1}{n}.
Step 1 applied to \dot P_m and \dot P_n. Thus (S(f;\dot P_m))_{m=1}^\infty is a Cauchy sequence in \mathbb{R}. By Theorem 3.5.5 (P06), it converges to a real number A := \lim_m S(f;\dot P_m). Letting m \to \infty in Step 2 gives \lvert S(f;\dot P_n) - A\rvert \le 1/n for all n \in \mathbb{N}.Cauchy Convergence Criterion for sequences in \mathbb{R} (P06). Given \varepsilon > 0, choose K \in \mathbb{N} with K > 2/\varepsilon. For any tagged partition \dot Q with \lVert Q\rVert < \delta_K, \lvert S(f;\dot Q) - A\rvert \le \lvert S(f;\dot Q) - S(f;\dot P_K)\rvert + \lvert S(f;\dot P_K) - A\rvert < \frac{1}{K} + \frac{1}{K} = \frac{2}{K} < \varepsilon.
Since \varepsilon > 0 is arbitrary, f \in \mathcal{R}[a,b] with \int_a^b f = A. ∎Triangle Inequality and Definition 7.1.1 with \delta_\varepsilon = \delta_K.   Marks: statement 1 · (\Rightarrow) proof 1.5 · (\Leftarrow) Cauchy sequence & limit A 2 · final 2/K < \varepsilon verification 1.5. </section>
<section class="page"> 24PYQ MAT-22 Q4also MAT-22 MCQ 17, MAT-21 MCQ 12, 16 · adapted Dirichlet's function and h(x) = x + 1 on \mathbb{Q}\cap[0,1], 0 on irrationals, are not in \mathcal{R}[0,1]. Know firstT11T10P14 SkeletonSet \varepsilon_0 = 1/2 (or 1). For any \eta > 0, pick a partition P of [0,1] with \lVert P\rVert < \eta; tag P with rationals to get \dot P and with irrationals to get \dot Q, and show \lvert S(\dot P) - S(\dot Q)\rvert \ge \varepsilon_0. Proofs (Bartle Example 7.2.2(b) & Exercise 7.2.2) Part 1: Dirichlet's function f(x) = 1 (x \in \mathbb{Q}\cap[0,1]), 0 (x \in [0,1]\setminus\mathbb{Q}) C7. Put \varepsilon_0 := \tfrac12 > 0, and let \eta > 0 be arbitrary. Choose any partition P = \{[x_{i-1},x_i]\}_{i=1}^n of [0,1] with \lVert P\rVert < \eta.Set \varepsilon_0 = 1/2 to negate Theorem 7.2.1. Since \mathbb{Q} and \mathbb{R}\setminus\mathbb{Q} are both dense in \mathbb{R}, each

subinterval [x_{i-1},x_i] contains a rational point t_i and an irrational point s_i. Let \dot P be P tagged with the rational points t_i, and let \dot Q be P tagged with the irrational points s_i.Density of rationals and irrationals (P14). Then S(f;\dot P) = \sum_{i=1}^n 1\cdot(x_i - x_{i-1}) = 1 and S(f;\dot Q) = \sum_{i=1}^n 0\cdot(x_i - x_{i-1}) = 0, so
\lvert S(f;\dot P) - S(f;\dot Q)\rvert = \lvert 1 - 0\rvert = 1 \ge \frac12 = \varepsilon_0.
Since \eta > 0 was arbitrary, the Cauchy Criterion 7.2.1 fails and f \notin \mathcal{R}[0,1] (MAT-22 MCQ 17 & MAT-21 MCQ 12 option B).Negation of Theorem 7.2.1. Part 2: h(x) = x + 1 (x \in \mathbb{Q}\cap[0,1]), 0 (x \in [0,1]\setminus\mathbb{Q}) (Bartle Ex 7.2.2; MAT-21 MCQ 16). Put \varepsilon_0 := 1 > 0. For any \eta > 0 and the same \dot P (rational tags t_i \in [0,1]) and \dot Q (irrational tags s_i), we have h(t_i) = t_i + 1 \ge 1 and h(s_i) = 0, so
S(h;\dot P) = \sum_{i=1}^n (t_i + 1)(x_i - x_{i-1}) \ge \sum_{i=1}^n 1\cdot(x_i - x_{i-1}) = 1, \qquad S(h;\dot Q) = 0.
Hence \lvert S(h;\dot P) - S(h;\dot Q)\rvert \ge 1 = \varepsilon_0, so h \notin \mathcal{R}[0,1] (MAT-21 MCQ 16 option B). ∎t_i \ge 0 \Rightarrow t_i + 1 \ge 1 on [0,1].   Marks: Cauchy negation setup 1 · Dirichlet proof 2.5 · h(x)=x+1 proof 2.5.
25PYQ MTS-26 Q18also MTS-22 Q6, MAT-21 Q7 State and prove the Squeeze Theorem (7.2.3); apply it to h(x) = x on [0,1] (Example 7.2.6(b)). Know firstT12T10T14P03 Theorem 7.2.3 · Squeeze Theorem (Bartle p. 209)Let f : [a,b] \to \mathbb{R}. Then f \in \mathcal{R}[a,b] if and only if for every \varepsilon > 0 there exist functions \alpha_\varepsilon and \omega_\varepsilon in \mathcal{R}[a,b] with
\alpha_\varepsilon(x) \le f(x) \le \omega_\varepsilon(x) \quad \text{for all } x \in [a,b] \qquad \text{and} \qquad \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) < \varepsilon.
 Proof of Theorem 7.2.3 & Example 7.2.6(b) (Bartle pp. 209–211) Proof of 7.2.3: (\Rightarrow) If f \in \mathcal{R}[a,b], take \alpha_\varepsilon = \omega_\varepsilon = f; then \int_a^b(\omega_\varepsilon - \alpha_\varepsilon) = 0 < \varepsilon. (\Leftarrow) Let \varepsilon > 0. Since \alpha_\varepsilon, \omega_\varepsilon \in \mathcal{R}[a,b], there exists \delta_\varepsilon > 0 such that any tagged partition \dot P with \lVert P\rVert < \delta_\varepsilon satisfies \big\lvert S(\alpha_\varepsilon;\dot P) - \int_a^b \alpha_\varepsilon\big\rvert < \varepsilon and \big\lvert S(\omega_\varepsilon;\dot P) - \int_a^b \omega_\varepsilon\big\rvert < \varepsilon.Definition 7.1.1 applied to \alpha_\varepsilon and \omega_\varepsilon, taking \min of the two deltas (M08). Since \alpha_\varepsilon \le f \le \omega_\varepsilon, we have S(\alpha_\varepsilon;\dot P) \le S(f;\dot P) \le S(\omega_\varepsilon;\dot P), and therefore \int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot P) < \int_a^b \omega_\varepsilon + \varepsilon.
If \dot Q is any other tagged partition with \lVert Q\rVert < \delta_\varepsilon, then \int_a^b \alpha_\varepsilon - \varepsilon < S(f;\dot Q) < \int_a^b \omega_\varepsilon + \varepsilon as well.Order preservation of Riemann sums (P02). Subtracting these two inequalities and using Theorem 7.1.5 gives
\lvert S(f;\dot P) - S(f;\dot Q)\rvert < \int_a^b \omega_\varepsilon - \int_a^b \alpha_\varepsilon +

2\varepsilon = \int_a^b (\omega_\varepsilon - \alpha_\varepsilon) + 2\varepsilon < \varepsilon + 2\varepsilon = 3\varepsilon.
Since \varepsilon > 0 is arbitrary, the Cauchy Criterion 7.2.1 implies f \in \mathcal{R}[a,b].Theorem 7.1.5 and Cauchy Criterion 7.2.1 (T10).
Application to h(x) = x on [0,1] (Example 7.2.6(b)): On the n-part equal partition P_n = (0, \tfrac1n, \tfrac2n, \dots, 1), define step functions \alpha_n(x) := \tfrac{k-1}{n} and \omega_n(x) := \tfrac{k}{n} for x \in [\tfrac{k-1}{n}, \tfrac{k}{n}) (k=1,\dots,n-1, and closed on [\tfrac{n-1}{n},1]). Then \alpha_n \le h \le \omega_n and by Theorem 7.2.5,
\int_0^1 \alpha_n = \frac{1}{n}\sum_{k=1}^n \frac{k-1}{n} = \frac{(n-1)n}{2n^2} = \frac12\left(1 - \frac1n\right), \quad \int_0^1 \omega_n = \frac{1}{n}\sum_{k=1}^n \frac{k}{n} = \frac{n(n+1)}{2n^2} = \frac12\left(1 + \frac1n\right).
Thus \int_0^1(\omega_n - \alpha_n) = \tfrac1n < \varepsilon for n > 1/\varepsilon, so h \in \mathcal{R}[0,1] by 7.2.3, and \tfrac12(1-\tfrac1n) \le \int_0^1 x\,dx \le \tfrac12(1+\tfrac1n)\ \forall n \implies \int_0^1 x\,dx = \tfrac12. ∎Sum of first n integers (P03) and Theorem 7.2.3.   \displaystyle\int_0^1 x\,dx = \frac12. Marks: statement of 7.2.3 1 · proof of 7.2.3 3 · step functions \alpha_n, \omega_n and \int_0^1 x\,dx = 1/2 2. </section>
<section class="page"> 26PYQ MTS-26 Q5also MAT-22 Q16 Every step function is integrable (Theorem 7.2.5); evaluate \int_0^3 g. Know firstT14T13T07M06 Theorem 7.2.5 · Integrability of Step Functions (Bartle p. 210)If \varphi : [a,b] \to \mathbb{R} is a step function, then \varphi \in \mathcal{R}[a,b]. If \varphi = \sum_{j=1}^m k_j\varphi_{J_j} where J_j has endpoints c_j < d_j, then \int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j). Proof of Theorem 7.2.5 (Bartle pp. 210–211) First, by Lemma 7.2.4, for any subinterval J \subseteq [a,b] with endpoints c < d, the elementary step function \varphi_J(x) = 1 on J and 0 elsewhere belongs to \mathcal{R}[a,b] with \int_a^b \varphi_J = d - c (for J = [c,d], \lVert P\rVert < \delta \Rightarrow (d-c)-2\delta \le S(\varphi_{[c,d]};\dot P) \le (d-c)+2\delta, so \delta_\varepsilon = \varepsilon/4 works; the other three interval types differ at \le 2 endpoints, so Theorem 7.1.3 applies).Lemma 7.2.4 (T13). By Definition 5.4.9, if \varphi : [a,b] \to \mathbb{R} is an arbitrary step function, [a,b] is the union of finitely many non-overlapping subintervals J_1, \dots, J_m with endpoints c_j \le d_j such that \varphi has constant value k_j on J_j. Thus \varphi can be written as the linear combination \varphi = \sum_{j=1}^m k_j\varphi_{J_j}.
Exercise 7.2.5 / M06. By Lemma 7.2.4 and linearity of the integral (Theorem 7.1.5(a),(b), applied inductively to m terms), \varphi \in \mathcal{R}[a,b] and
\int_a^b \varphi = \sum_{j=1}^m k_j\int_a^b \varphi_{J_j} = \sum_{j=1}^m k_j(d_j - c_j). \quad \blacksquare
Theorem 7.1.5(a),(b) (T07).   Example 7.2.6(a): For g(x) = 2 on [0,1] and g(x) = 3 on (1,3], we have g = 2\varphi_{[0,1]} + 3\varphi_{(1,3]}, so \int_0^3 g = 2(1 - 0) + 3(3 - 1) = 2 + 6 = 8. \displaystyle\int_a^b \varphi = \sum_{j=1}^m k_j(d_j - c_j) and \displaystyle\int_0^3

g = 8. Marks: Lemma 7.2.4 proof/statement 2 · decomposition \varphi = \sum k_j\varphi_{J_j} 1.5 · linearity & formula 1.5 · \int_0^3 g = 8 1.
27PYQ MAT-22 Q37b Restriction to [c,d] \subseteq [a,b] (Corollary 7.2.10) and subdivision formula (Corollary 7.2.11). Know firstT18T17 Corollaries 7.2.10 & 7.2.11 (Bartle p. 214)If f \in \mathcal{R}[a,b] and [c,d] \subseteq [a,b], then f|_{[c,d]} \in \mathcal{R}[c,d]. If a = c_0 < c_1 < \cdots < c_m = b, then \int_a^b f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f. Proofs Proof of Corollary 7.2.10: Let f \in \mathcal{R}[a,b] and a \le c < d \le b. If c = a, then f|_{[c,b]} = f \in \mathcal{R}[c,b] by hypothesis; if c > a, then c \in (a,b), so the Additivity Theorem 7.2.9 (\Rightarrow) implies that the restriction f|_{[c,b]} belongs to \mathcal{R}[c,b].Additivity Theorem 7.2.9 (\Rightarrow) at the cut point c \in (a,b). Next, consider f|_{[c,b]} \in \mathcal{R}[c,b]. If d = b, then f|_{[c,d]} = f|_{[c,b]} \in \mathcal{R}[c,d]; if d < b, then d \in (c,b), so applying Theorem 7.2.9 (\Rightarrow) to f|_{[c,b]} on [c,b] at the interior point d shows that its restriction f|_{[c,d]} belongs to \mathcal{R}[c,d].Second application of Theorem 7.2.9 (\Rightarrow) at d \in (c,b). Proof of Corollary 7.2.11 (by induction on m): By Corollary 7.2.10, each restriction f|_{[c_{i-1},c_i]} belongs to \mathcal{R}[c_{i-1},c_i]. For m = 1, \int_a^b f = \int_{c_0}^{c_1} f is immediate; for m = 2, \int_a^b f = \int_{c_0}^{c_1} f + \int_{c_1}^{c_2} f is Theorem 7.2.9. Assume the formula holds for m - 1 subintervals. Given a = c_0 < c_1 < \cdots < c_{m-1} < c_m = b, apply Theorem 7.2.9 at c = c_{m-1} \in (a,b) to get \int_a^b f = \int_a^{c_{m-1}} f + \int_{c_{m-1}}^{c_m} f = \sum_{i=1}^{m-1} \int_{c_{i-1}}^{c_i} f + \int_{c_{m-1}}^{c_m} f = \sum_{i=1}^m \int_{c_{i-1}}^{c_i} f. \quad \blacksquare Induction hypothesis applied on [a,c_{m-1}].   Marks: split at c 1.5 · split at d 1.5 · induction base & step for 7.2.11 3.
28PYQ MAT-22 MCQ 24, 25 · adaptedalso MAT-21 MCQ 18 · adapted Bounded f integrable on every [c,b] (c \in (a,b)) is in \mathcal{R}[a,b] (Bartle Ex 7.2.11–7.2.13); application to \sin(1/x). Know firstT18T12T17T08 Proof of Bartle Exercises 7.2.11–7.2.13 Assume \lvert f(x)\rvert \le M on [a,b] (M > 0) and f|_{[c,b]} \in \mathcal{R}[c,b] for every c \in (a,b). Given \varepsilon > 0, choose c \in (a,b) such that c - a < \dfrac{\varepsilon}{4M}.Make the trouble strip [a,c] near a narrower than \varepsilon/(4M). Define \alpha_c, \omega_c : [a,b] \to \mathbb{R} by \alpha_c(x) := -M and \omega_c(x) := M for x \in [a,c), and \alpha_c(x) := \omega_c(x) := f(x) for x \in [c,b]. Since \alpha_c, \omega_c are constant on [a,c) and equal to f \in \mathcal{R}[c,b] on [c,b], Theorem 7.2.9 (\Leftarrow) and Theorem 7.1.3 imply \alpha_c, \omega_c \in \mathcal{R}[a,b] and \alpha_c(x) \le f(x) \le \omega_c(x) on [a,b].Additivity Theorem 7.2.9 (\Leftarrow) glues the constant piece on [a,c] to f on [c,b]. Since \omega_c - \alpha_c = 2M on [a,c) and 0 on [c,b], we have
\int_a^b (\omega_c - \alpha_c) = \int_a^c 2M + \int_c^b 0 = 2M(c - a) < 2M\cdot\frac{\varepsilon}{4M} = \frac{\varepsilon}{2} < \varepsilon.
By the Squeeze Theorem 7.2.3, f \in \mathcal{R}[a,b].Squeeze Theorem

7.2.3 (T12). Furthermore, for any c \in (a,b), Theorem 7.2.9 gives \int_a^b f - \int_c^b f = \int_a^c f, so \left\lvert\int_a^b f - \int_c^b f\right\rvert = \left\lvert\int_a^c f\right\rvert \le M(c - a) \to 0 as c \to a^+, proving \lim_{c\to a^+} \int_c^b f = \int_a^b f. (Thus in MAT-22 MCQ 24 / MAT-21 MCQ 18, the statement that does not hold is (B) f \notin \mathcal{R}[a,b].)Exercise 7.1.8 bound \lvert\int_a^c f\rvert \le M(c-a) and Squeeze Theorem for limits. Application to g(x) = \sin(1/x) (x \in (0,1]), g(0) = 0 (Ex 7.2.12, MAT-22 MCQ 25): We have \lvert g(x)\rvert \le 1 on [0,1], and for each c \in (0,1), g is continuous on [c,1] and hence in \mathcal{R}[c,1] by Theorem 7.2.7. By Step 3, g \in \mathcal{R}[0,1] (MCQ 25 option A).Apply Exercise 7.2.11 with M = 1. Why boundedness cannot be dropped (Ex 7.2.13): f(x) = 1/x on (0,1], f(0) = 0 C16 is continuous on [c,1] (so f \in \mathcal{R}[c,1]) for every c \in (0,1), but f is unbounded on [0,1], so f \notin \mathcal{R}[0,1] by Theorem 7.1.6. ∎Counterexample C16.   Marks: \alpha_c, \omega_c construction & Squeeze 2.5 · limit \int_c^b f \to \int_a^b f 1.5 · \sin(1/x) \in \mathcal{R}[0,1] 1 · 1/x counterexample 1. </section>
=== NOTES / DOUBTS (STAGE 5 OF 6) ===

- Completed Stage 5 (src/m3/answers_1.html and src/m3/answers_2.html): formal solutions to Section A (Q1–Q20) and the first half of Section B (Q21–Q28), with restatements, Know first chips, skeletons, step-by-step proofs with .why justifications, .final answer boxes, .marking splits, and one-line explanations of ambiguous past MCQs (MTS-22 MCQ 7, MAT-22 MCQ 18, MAT-22 MCQ 24).
- Final stage (Stage 6) is next: src/m3/answers_3.html (solutions to Q29–Q40) and src/m3/figs_m3.py (all 18 SVG figure functions F01_...–F18_...).
- **Action needed for src/m3/figs_m3.py in Stage 6:** Please paste or attach figs.py and figs_m2.py in your next message so src/m3/figs_m3.py can use the exact S and Plot class definitions, colour constants, and helper functions (sub(), sup(), etc.).$]